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Final Exam

Program: Computer Science
Course Name: Advanced Data Structures
Course Code: CS316
Instructor(s): Dr. Amin Allam

Date: 18/1/2022
Duration: 2 hours
Total Marks: 60 marks

تعليمات هامة:

- حيازة التليفون المحمول مفتوحا داخل لجنة الامتحان يعتبر حالة غش تستوجب العقاب وإذا كان ضروري الدخول بالمحمول فيوضع مغلقا في الحقائب.
- لا يسمح بدخول سماعة الأذن أو البلوتوث.
- لا يسمح بدخول أي كتب أو ملازم أو أوراق داخل اللجنة والمخالفة تعتبر حالة غش.

- Exam consists of 40 multiple-choice questions in 6 pages. Each question weights 1.5 marks.
- Record in the bubble sheet exactly ONE answer for each question.

Qa ⇒ For questions 1 to 6, consider applying a sequence of operations on a B-tree of order 5 which is initially-empty. The maximum number of keys per node is 4. The minimum number of keys per node (except root) is 2.

1 Insert(1), Insert(2), Insert(3), Insert(5), Insert(6).

Applying the above operations on empty B-tree of order 5, one node contains exactly and only the keys:

☐ A 1 ☒ B 1-2 ☐ C 1-2-3 ☐ D 1-2-3-5 ☐ E 1-2-3-5-6

2 Insert(1), Insert(2), Insert(3), Insert(5), Insert(6), Insert(8), Insert(9).

Applying the above operations on empty B-tree of order 5, one node contains exactly and only the keys:

☐ A 1-2-3 ☐ B 2-3 ☒ C 3 ☐ D 3-5 ☐ E 3-5-6

3 Insert(1), Insert(2), Insert(3), Insert(5), Insert(6), Insert(8), Insert(9), Insert(4), Insert(7).

Applying the above operations on empty B-tree of order 5, one node contains exactly and only the keys:

☐ A 1-2-4 ☐ B 2-4-5 ☐ C 4-5 ☒ D 4-5-7 ☐ E 1-2-4-5

4 Insert(1), Insert(2), Insert(3), Insert(5), Insert(6), Insert(8), Insert(9), Insert(4), Insert(7), Delete(1).

Applying the above operations on empty B-tree of order 5, one node contains exactly and only the keys:

☐ A 9 ☒ B 8-9 ☐ C 7-8-9 ☐ D 5-7-8-9 ☐ E 5-9

5 Insert(1), Insert(2), Insert(3), Insert(5), Insert(6), Insert(8), Insert(9), Insert(4), Insert(7), Delete(1), Delete(5), Delete(4), Delete(3).

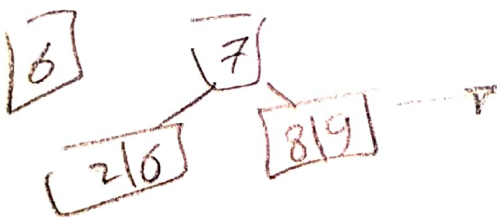
Applying the above operations on empty B-tree of order 5, one node contains exactly and only the keys:

☐ A 2 ☒ B 2-6 ☐ C 2-6-7 ☐ D 2-6-7-8 ☐ E 2-7

6 Insert(1), Insert(2), Insert(3), Insert(5), Insert(6), Insert(8), Insert(9), Insert(4), Insert(7), Delete(1), Delete(5), Delete(4), Delete(3), Delete(8).

Applying the above operations on empty B-tree of order 5, one node contains exactly and only the keys:

☐ A 2 ☐ B 2-6 ☒ C 2-6-7 ☐ D 2-6-7-9 ☐ E 2-7



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Qb ⇒ For questions 7 to 12, consider the suffix tree of the string $abtababtbtabt\$$. The distance between two nodes is the number of edges on the simple path connecting these two nodes:

7 The suffix tree of string $abtababtbtabt\$$ contains the following number of nodes:
☐ A 18 ☐ B 19 ☐ C 20 ☐ D 21 ☒ E 22

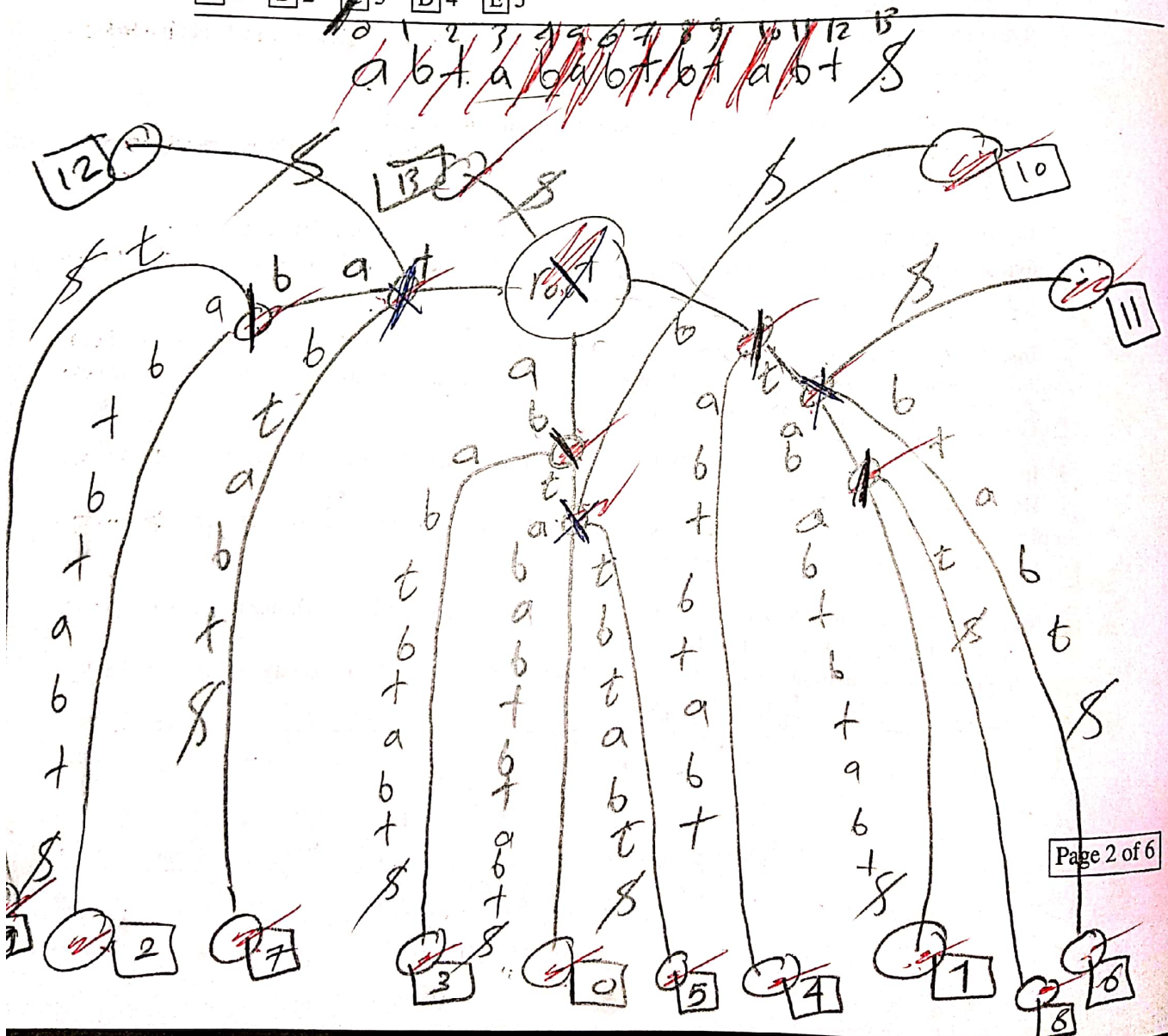
8 In the suffix tree of string $abtababtbtabt\$$, the number of nodes having 2 children is:
☐ A 0 ☐ B 1 ☐ C 2 ☐ D 3 ☒ E 4

9 In the suffix tree of string $abtababtbtabt\$$, the number of nodes having at least 3 children is:
☐ A 0 ☐ B 1 ☐ C 2 ☐ D 3 ☒ E 4

10 In the suffix tree of string $abtababtbtabt\$$, the distance between the root and the leaf containing the ID 3 of the suffix $ababtbtabt\$$:
☐ A 1 ☒ B 2 ☐ C 3 ☐ D 4 ☐ E 5

11 In the suffix tree of string $abtababtbtabt\$$, the distance between the root and the leaf containing the ID 6 of the suffix $btbtabt\$$:
☐ A 1 ☐ B 2 ☒ C 3 ☐ D 4 ☐ E 5

12 In the suffix tree of string $abtababtbtabt\$$, the distance between the root and the leaf containing the ID 9 of the suffix $tabt\$$:
☐ A 1 ☐ B 2 ☒ C 3 ☐ D 4 ☐ E 5



Qc ⇒ For questions 13 to 18, consider the construction of the suffix array of the string abtababtabt\$ using the prefix doubling algorithm. At iteration 0 (the first iteration), the initial intermediate array is {1 2 3 1 2 1 2 3 1 2 3 0} whose maximum integer is 3 and whose sum of integers is 21.

13 At iteration 1 (the second iteration) of the prefix doubling algorithm for constructing the suffix array of the string abtababtabt\$, the maximum integer in the intermediate array is:

- A 3 B 4 ☒ C 5 D 6 E 7

14 At iteration 1 (the second iteration) of the prefix doubling algorithm for constructing the suffix array of the string abtababtabt\$, the sum of integers in the intermediate array is:

- ☒ A 29 B 32 C 35 D 37 E 39

15 At iteration 2 (the third iteration) of the prefix doubling algorithm for constructing the suffix array of the string abtababtabt\$, the maximum integer in the intermediate array is:

- A 6 B 7 C 8 ☒ D 9 E 10

16 At iteration 2 (the third iteration) of the prefix doubling algorithm for constructing the suffix array of the string abtababtabt\$, the sum of integers in the intermediate array is:

- A 45 B 47 C 49 D 52 ☒ E 54

17 At iteration 3 (the fourth iteration) of the prefix doubling algorithm for constructing the suffix array of the string abtababtabt\$, the maximum integer in the intermediate array is:

- A 7 B 8 C 9 D 10 ☒ E 11

18 At iteration 3 (the fourth iteration) of the prefix doubling algorithm for constructing the suffix array of the string abtababtabt\$, the sum of integers in the intermediate array is:

- A 50 B 54 C 60 ☒ D 66 E 70

1 2 3 4 5 6 7 8 9 10 11
abtababtabt\$

Index	0	1	2	3	4	5	6	7	8	9	10	11
iter	0	1	2	3	4	5	6	7	8	9	10	11
0	1	2	3	1	2	1	2	3	1	2	3	0
1	1	3	5	1	2	1	3	5	1	3	4	0
2	3	6	8	1	4	3	6	9	2	5	7	0
3	3	7	10	1	5	4	8	11	2	6	9	0

1 2 3 1 2 1 2 3 1 2 3 0 → ①
 1 3 5 1 2 1 3 5 1 3 4 0 → ②
 1 3 5 1 4 3 6 9 2 5 7 0 → ③
 3 6 8 1 5 4 8 11 2 6 9 0 → ④

(15) max

2, 3, 4, 6, 8, 9, 10, 12

Qd ⇒ For questions 19 to 24, consider the static segment tree constructed over these 7 input intervals: $[2,9]$ $[3,6]$ $[3,8]$ $[4,12]$ $[8,12]$ $[9,15]$ $[10,12]$ $[10,15]$

(15) max

19 The canonical representation of the interval $[4,12]$ with respect to the above segment tree is:

- A $[4,12]$ B $[4,6]$ $[6,8]$ $[8,12]$ C $[4,8]$ $[8,12]$ D $[4,10]$ $[10,12]$ E $[4,8]$ $[8,10]$ $[10,12]$

20 The canonical representation of the interval $[2,15]$ with respect to the above segment tree is:

- A $[2,15]$ B $[2,4]$ $[4,8]$ $[8,15]$ C $[2,8]$ $[8,15]$ D $[2,4]$ $[4,8]$ $[8,15]$ E $[2,10]$ $[10,15]$

21 The canonical representation of the interval $[6,12]$ with respect to the above segment tree is:

- A $[6,12]$ B $[6,8]$ $[8,10]$ $[10,12]$ C $[6,8]$ $[8,12]$ D $[6,10]$ $[10,12]$ E $[6,9]$ $[9,10]$ $[10,12]$

22 The canonical representation of the interval $[4,10]$ with respect to the above segment tree is:

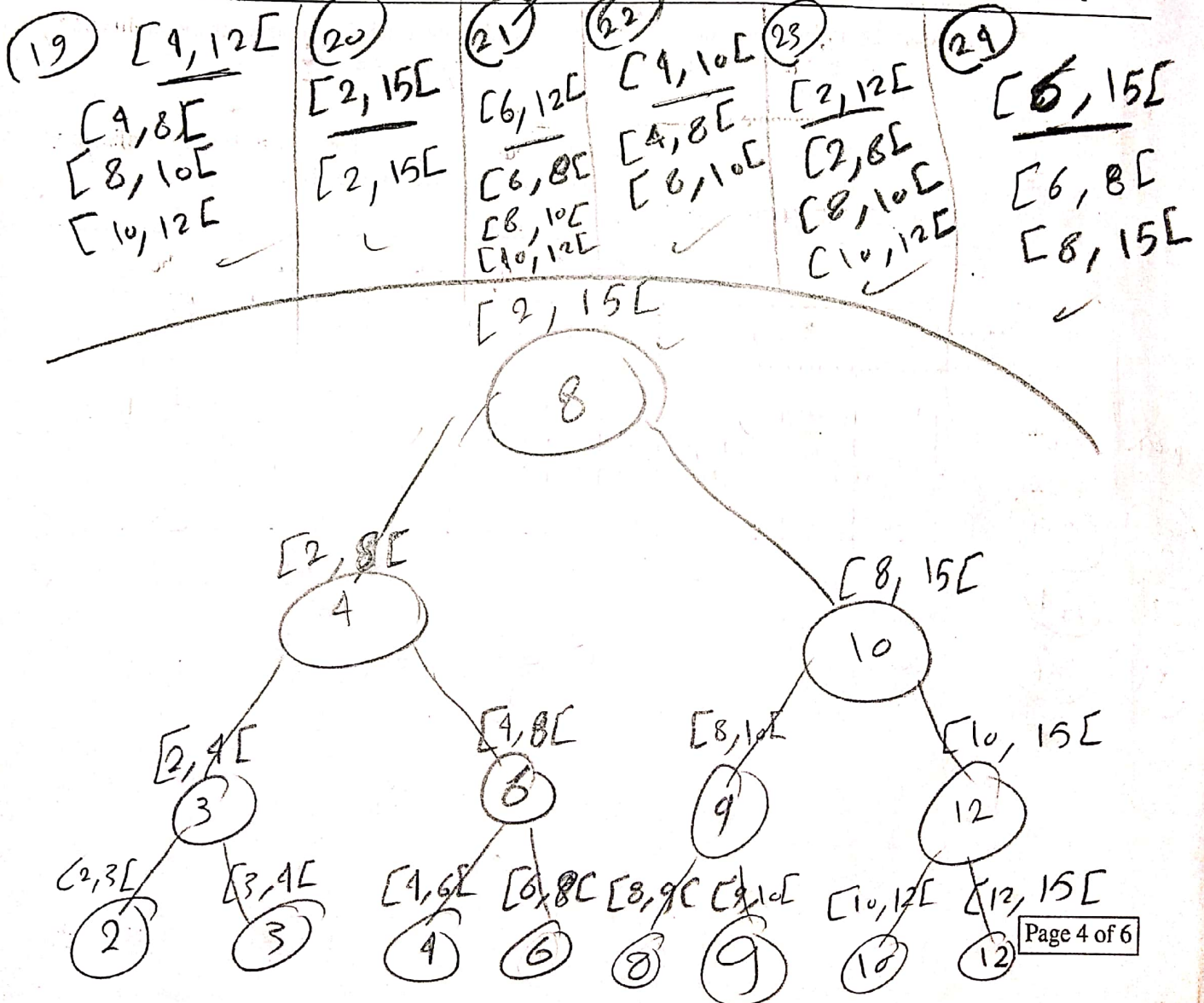
- A $[4,10]$ B $[4,6]$ $[6,8]$ $[8,10]$ C $[4,8]$ $[8,10]$ D $[4,8]$ $[8,9]$ $[9,10]$ E $[4,9]$ $[9,10]$

23 The canonical representation of the interval $[2,12]$ with respect to the above segment tree is:

- A $[2,12]$ B $[2,6]$ $[6,9]$ $[9,12]$ C $[2,8]$ $[8,10]$ $[10,12]$ D $[2,10]$ $[10,12]$ E $[2,8]$ $[8,12]$

24 The canonical representation of the interval $[6,15]$ with respect to the above segment tree is:

- A $[6,15]$ B $[6,8]$ $[8,10]$ $[10,15]$ C $[6,8]$ $[8,15]$ D $[6,12]$ $[12,15]$ E $[6,10]$ $[10,15]$



Left $(1,3) \rightarrow (2,6) \rightarrow (3,5) \rightarrow (4,2) \rightarrow (1,3) \mid (3,5) \mid (2,6)$
 Right $(5,8) \rightarrow (6,7) \rightarrow (7,1) \rightarrow (8,4) \rightarrow (7,1) \mid (6,7) \mid (5,8)$

Qe $\Rightarrow$ For questions 25 to 30, consider the static Kd tree constructed over these 8 two dimensional points: $(1,3), (2,6), (3,5), (4,2), (5,8), (6,7), (7,1), (8,4)$. The root node (first level) contains the separating vertical line $X=4.5$. The second level nodes contain the separating horizontal lines $Y=4.0, Y=5.5$. The third level nodes contain the separating vertical lines $X=2.5, X=2.5, X=7.5, X=5.5$. The Kd tree enables finding all points residing inside a query rectangle efficiently. A subset of the above points are tested for existence inside the query rectangle. All the following questions ask for the tested points, regardless of whether they are actually inside the query rectangle or not.

25 When the query rectangle $[x_{min}=0.5, x_{max}=4.2] [y_{min}=4.2, y_{max}=8.2]$ is applied to the above Kd tree, exactly and only the following points will be tested for existence in the query rectangle:

- A $(2,6), (3,5)$ B $(1,3), (2,6), (3,5)$ C $(4,2), (2,6), (3,5)$ D $(1,3), (4,2), (2,6), (3,5)$
 E $(1,3), (4,2), (2,6), (3,5), (7,1)$

26 When the query rectangle $[x_{min}=0.5, x_{max}=4.2] [y_{min}=3.2, y_{max}=8.2]$ is applied to the above Kd tree, exactly and only the following points will be tested for existence in the query rectangle:

- A $(2,6), (3,5)$ B $(1,3), (2,6), (3,5)$ C $(4,2), (2,6), (3,5)$ D $(1,3), (4,2), (2,6), (3,5)$
 E $(1,3), (4,2), (2,6), (3,5), (7,1)$

27 When the query rectangle $[x_{min}=4.8, x_{max}=8.2] [y_{min}=1.5, y_{max}=4.5]$ is applied to the above Kd tree, exactly and only the following points will be tested for existence in the query rectangle:

- A $(7,1), (8,4)$ B $(7,1), (8,4), (5,8)$ C $(7,1), (8,4), (6,7)$ D $(7,1), (8,4), (5,8), (6,7)$
 E $(7,1), (8,4), (5,8), (6,7), (4,2)$

28 When the query rectangle $[x_{min}=4.8, x_{max}=8.2] [y_{min}=1.5, y_{max}=8.2]$ is applied to the above Kd tree, exactly and only the following points will be tested for existence in the query rectangle:

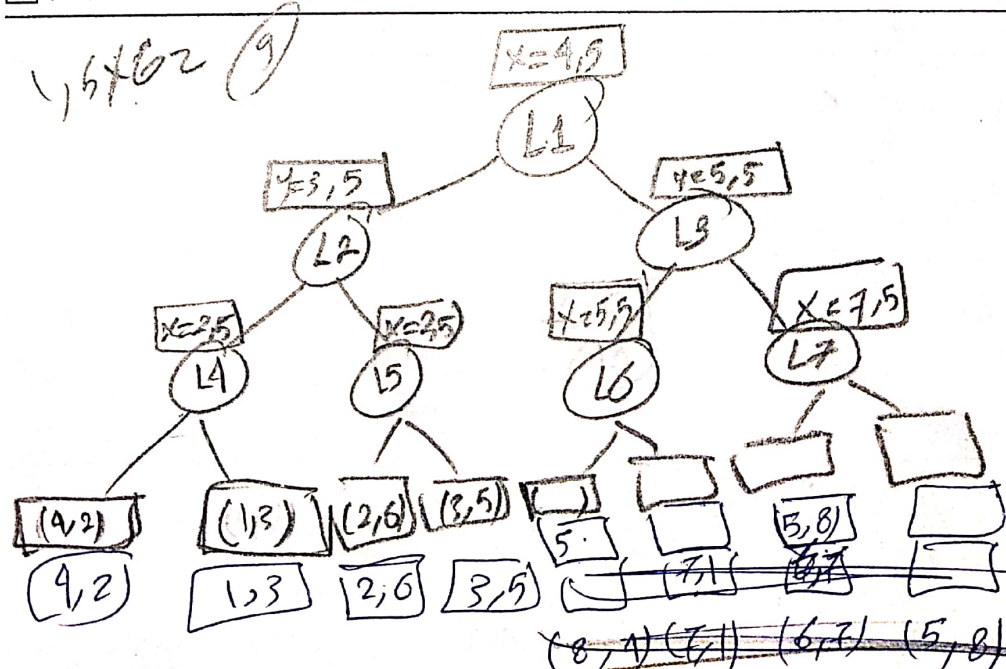
- A $(7,1), (8,4)$ B $(7,1), (8,4), (5,8)$ C $(7,1), (8,4), (6,7)$ D $(7,1), (8,4), (5,8), (6,7)$
 E $(7,1), (8,4), (5,8), (6,7), (4,2)$

29 When the query rectangle $[x_{min}=2.8, x_{max}=6.5] [y_{min}=2.8, y_{max}=5.2]$ is applied to the above Kd tree, exactly and only the following points will be tested for existence in the query rectangle:

- A $(4,2), (3,5)$ B $(4,2), (3,5), (7,1)$ C $(4,2), (2,6), (3,5), (7,1)$ D $(1,3), (4,2), (2,6), (3,5)$
 E $(1,3), (4,2), (2,6), (3,5), (7,1)$

30 When the query rectangle $[x_{min}=1.5, x_{max}=6.5] [y_{min}=2.8, y_{max}=5.2]$ is applied to the above Kd tree, exactly and only the following points will be tested for existence in the query rectangle:

- A $(4,2), (3,5)$ B $(4,2), (3,5), (7,1)$ C $(4,2), (2,6), (3,5), (7,1)$ D $(1,3), (4,2), (2,6), (3,5)$
 E $(1,3), (4,2), (2,6), (3,5), (7,1)$



Qf 31 Assume disjoint sets of 5 elements returns: Find(0)=0, Find(1)=0, Find(2)=2, Find(3)=4, Find(4)=4. After performing Union(1,3) to union the sets containing element 1 and element 3 using the weight rule, Find(2) will return:

- A 0 B 1 ☒ C 2 D 3 E 4

$$P(0)=0, P(1)=0$$

32 Assume disjoint sets of 5 elements returns: Find(0)=0, Find(1)=2, Find(2)=4, Find(3)=4, Find(4)=4. After performing Union(1,2) to union the sets containing element 1 and element 2 using the weight rule, Find(4) will return:

- A 0 B 1 C 2 D 3 ☒ E 4

$$P(0)=0, P(1)=0$$

33 Assume disjoint sets of 5 elements returns: Find(0)=0, Find(1)=0, Find(2)=4, Find(3)=4, Find(4)=4. After performing Union(1,2) to union the sets containing element 1 and element 2 using the weight rule, Find(0) will return:

- A 0 B 1 C 2 D 3 ☒ E 4

34 Bloom filters have the following disadvantage compared to hash tables:

- A produce true positives B produce true negatives ☒ C produce false positives
D produce false negatives E consume less space

False positive

35 The following structure allows efficient searching for substrings of an indexed string:

- A interval tree B segment tree C range tree D Kd tree ☒ E suffix tree

36 The following structure allows efficient finding of all input points belonging to a query interval:

- A interval tree B segment tree ☒ C range tree D suffix tree E suffix array

37 The following structure allows efficient finding of all input intervals overlapping with a query interval. Also, it allows efficient insertion and removal of input intervals:

- ☒ A interval tree B segment tree C range tree D Kd tree E suffix tree

38 Using orthogonal 3d range tree storing n points, the time complexity of searching for all input points belonging to a query rectangle is: (assuming that there are k points satisfying the query):

- A $O(k + \log^2 n)$ B $O(\log^2(n + k))$ ☒ C $O(k + \log^3 n)$ D $O(k^3 + \log^3 n)$ E $O(k \log^3 n)$

39 The space complexity of orthogonal 3d range tree storing n points is:

- A $O(n + \log n)$ B $O(n^3 + \log^3 n)$ C $O(n \log^3 n)$ ☒ D $O(n \log^2 n)$ E $O(n^3)$

40 The space complexity of 3d Kd tree storing n points is:

- ☒ A $O(n)$ B $O(n^2)$ C $O(n^3)$ D $O(n \log n)$ E $O(n^2 \log n)$